Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
	(b)	(i)		Light intensity / distance of the torch from the LDR	1			1		1
		(ii)		Resistance of the LDR accept voltmeter reading/ ammeter reading /meter readings	1			1		1
		(iii)		Total resistance = 2.5 + 1 = 3.5 kΩ (1) (sub) $I = \frac{9}{3500}$ (1)(manip and conversion) $=2.57 \times 10^{-3}$ / 0.00257 (1) V = 0.00257 (ecf) × 2500 (1) (sub) =6.42 V (1) Or Total resistance = 2.5 + 1 = 3.5 kΩ (1) (sub) PD across LDR + 1 kohm = 9 V (1) PD across LDR 2.5/3.5 × 9 = 6.43V (1) $I = \frac{6.43\ (ecf)}{2500}$ (1) (manip and conversion) =0.00257 (1) If conversion not done, expect answers of 2.57 A and 6428/9 V for (4) Do not penalise incorrect rounding	2	3		5	5	5
		(iv)		Keep light intensity constant (1) Add a variable resistor / use variable power supply (1) Take positive and negative readings of current and voltage (1)			3	3		3
				Question 7 total	**6**	**7**	**3**	**16**	**4**	**14**